UNIT IV

Metal Matrix Composites (MMCs)

Reinforcements & base metals | need & applications | fabrication processes | properties

I. Introduction to Metal Matrix Composites

A **Metal Matrix Composite (MMC)** consists of a **metal or alloy matrix** (Al, Mg, Ti, Cu, Ni ...) reinforced by a **hard, stiff, strong second phase** (ceramic fibres, whiskers or particles; sometimes a second metal). The metal gives ductility, toughness and conductivity; the reinforcement gives stiffness, strength, wear and creep resistance and low thermal expansion.

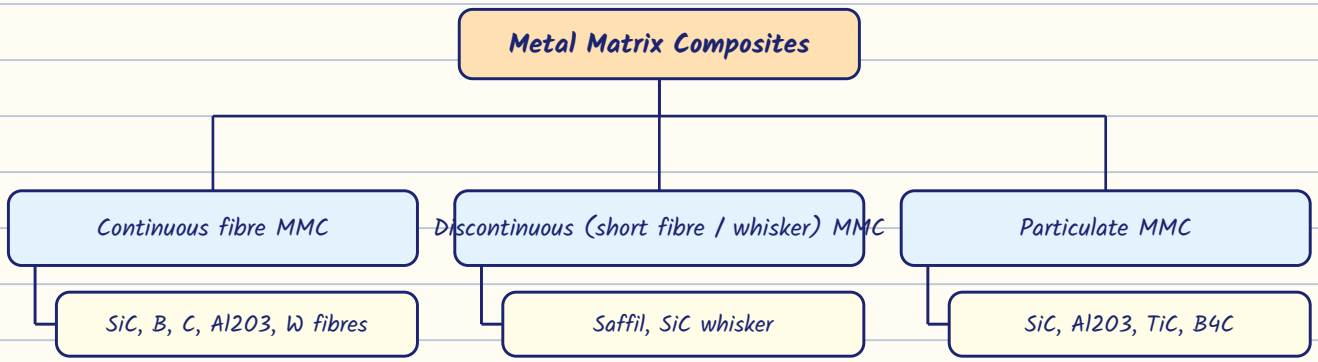


Fig. 1 Classification of MMCs by reinforcement type

I.1 Advantages over monolithic metals and polymer matrix composites

- **High specific strength and stiffness**, tailored thermal expansion (CTE) - matches Si, GaAs and ceramics.
- **Higher service temperature** than PMCs (Al: 300 °C, Ti: 600 °C, Ni: > 1000 °C) and retention of properties at heat.
- Excellent **wear and abrasion resistance**, fatigue and creep resistance; high thermal and electrical conductivity.
- **No moisture absorption, no out-gassing**, non-flammable, resistant to radiation and UV; better transverse and shear strength than PMCs.
- Can be machined, joined, heat-treated by metal-working methods (extrusion, forging, rolling).

I.2 Limitations

- **High cost** of reinforcements and processing; heavier than PMCs ($\rho \approx 2.8$-4.5 g/cc).

- **Reinforcement – matrix reaction** at high temperature (Al_4C_3 in C/Al; brittle phases at interface); poor wetting of ceramics by molten metal.
- Reduced ductility, fracture toughness; **difficult to machine** (tool wear) and to recycle; galvanic corrosion at the interface.

II. Reinforcement Materials

Reinforcements are selected for high strength / modulus, low density, **chemical compatibility** and **wettability** with the matrix, thermal stability, appropriate CTE and cost.

2.1 Types and characteristics

Type	Examples	Typical properties	Remark
Continuous fibres	Carbon (graphite), boron, SiC (SCS-6, Nicalon), Al_2O_3 (Nextel, FP), tungsten, steel wire	Diameter 8–140 µm; E = 200–600 GPa; σ = 2–4 GPa	Highest, anisotropic properties; costly; need coating
Short fibres / whiskers	Saffil (δ-Al_2O_3), SiC whiskers, K-titanate, Kaowool	d = 0.1–10 µm, L/d = 10–100; E = 300–500 GPa	Near isotropic; health hazard of whiskers
Particulates	SiC, Al_2O_3, B_4C, TiC, TiB_2, WC, graphite, fly ash, Si_3N_4	Size 1–100 µm; hardness 20–35 GPa	Cheapest, isotropic, easy casting and PM; most widely used
Metal wire	Mo, W, steel, Nb	High density, very ductile	Used in Cu, Ni, Al matrices

Reinforcement	Density (g/cc)	E (GPa)	CTE (10^{-6}/K)	Strength (MPa)
SiC	3.2	400–480	4.3	2000–3000 (fibre / whisker)
Al_2O_3	3.9	350–400	7–8	1500–2000
B_4C	2.5	450	5.0	~ 3000
Carbon (HM)	1.9–2.2	400–600	−1 to 0	2500–3500
Boron	2.6	400	5.0	3500
TiC	4.9	440	7.4	> 1500

2.2 Selection criteria for reinforcement

- **High modulus & strength**, low density → good specific properties.
- **Chemically stable** with the matrix at fabrication and service temperature (reaction layer must be thin, < 100 nm).
- **Wettability** by molten metal (Mg or Ti additions improve wetting of SiC / Al_2O_3 by Al); coatings (SiC, TiB_2, C, Ni) on fibres.
- **CTE mismatch** small to avoid residual stresses and thermal fatigue; **cost** and availability; safety (whisker toxicity).

III. Base Metals (Matrix) – Types and Selection

Matrix	Density (g/cc)	Max. use temp.	Reinforcements used	Applications
Aluminium (6061, 2124, A356)	2.7	≈ 300 °C	SiC, Al_2O_3, B_4C, C, B	Pistons, brake discs, aerospace struts, electronics
Magnesium (AZ91, ZK60)	1.74	≈ 250 °C	SiC, C, Al_2O_3	Lightest MMC: aerospace, auto
Titanium (Ti-6Al-4V)	4.5	≈ 600 °C	SiC fibre, TiC, TiB, B_4C	Jet-engine, aerospace skin
Copper	8.9	≈ 400 °C	Graphite, W, Mo, SiC, Al_2O_3	Heat sinks, electrical contacts
Nickel / superalloys	8.2–8.9	> 1000 °C	W, SiC, Al_2O_3, TaC	Turbine blades, hot sections
Iron / steel	7.8	≈ 600 °C	TiC, WC, Al_2O_3	Wear tools, dies
Zinc, Lead, Tin	7.1 / 11.3 / 7.3	low	Graphite, SiC	Bearings, low-temperature

Criteria for selecting the base metal

> **Service temperature** and environment (oxidation, corrosion).
> **Density / specific strength** required (Mg < Al < Ti < Cu < Ni).
> **Compatibility and wetting** with reinforcement; reaction products.
> **Processability**: castability, formability, joinability, heat-treatability (Al 6061-T6, 2124).
> **Cost and availability**; conductivity requirement (Cu, Al for thermal management).

IV. Need for Production of MMCs and Applications

4.1 Why MMCs are needed

- Monolithic alloys cannot give **high specific stiffness, wear resistance, low CTE and high-temperature strength together**.
- Demand for **lightweight** fuel-efficient vehicles and aircraft (weight saving 20-40 %).
- **Thermal management** of electronics: heat must be removed while CTE matches the chip (Al-SiC, Cu-W).
- Replace **cast iron** in brakes and cylinder liners (lighter, better wear), and expensive Ti / Ni in some hot parts.
- Better **creep, fatigue and dimensional stability** for space structures.

4.2 Applications

Sector	Components (MMC used)
Automobile	Pistons, piston crowns, cylinder liners, connecting rods, brake rotors / drums (Al-SiC), drive shaft (Al-SiC), engine blocks, calipers
Aerospace & defence	Space-shuttle orbiter mid-fuselage struts (B/Al), satellite structures (Gr/Mg), fan-exit guide vanes, rotor blades, missile fins, helicopter blade sleeves, armour
Electronics & thermal	Microprocessor heat sinks / substrates, IGBT base plates, RF packages (Al-SiC, Cu-W, Cu-Mo, Al-Gr)
Sports & recreation	Bicycle frames, golf clubs, tennis rackets (Al-Al_2O_3)
Industrial	Cutting tools (WC-Co), dies, pump parts, rollers, bearings, wear plates

V. Fabrication Processes for MMCs

MMC production methods are grouped as **solid-state (powder metallurgy, diffusion bonding)**, **liquid-state (stir casting, infiltration, squeeze casting, spray deposition)**, **vapour-state** and **in-situ** processes.

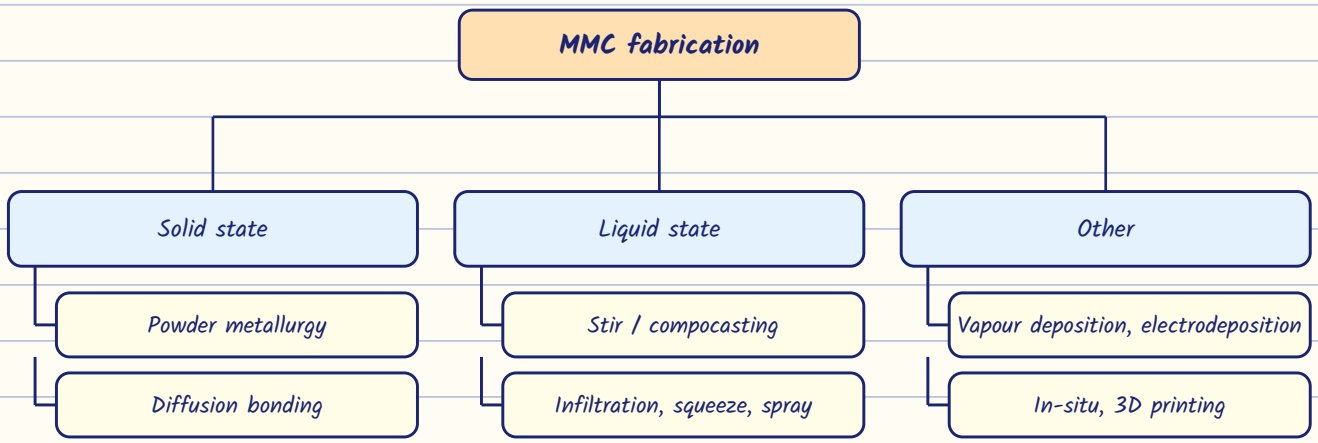


Fig. 2 Routes for manufacturing MMCs

5.1 Powder metallurgy (PM) technique

Metal powder (10-50 μm) and reinforcement (particles / short fibres) are **blended**, **compacted** and **consolidated** below the melting point of the matrix - no melt, so no reaction, and high V_f (up to 50 %) is possible.

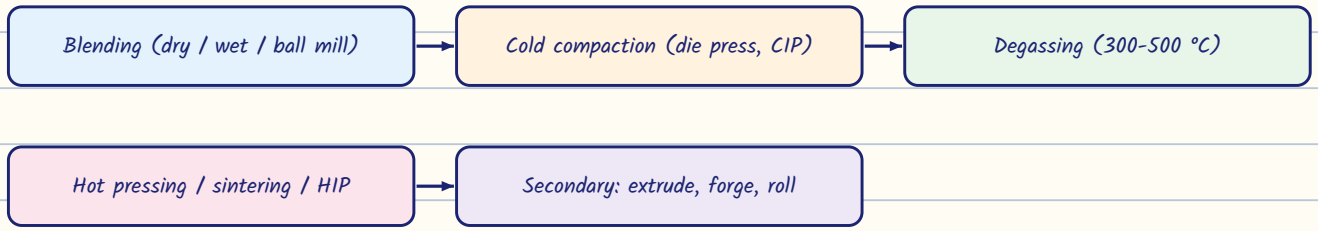


Fig. 3 Powder metallurgy route

- **Blending**: uniform mixing; ball-milling / mechanical alloying breaks agglomerates.
- **Cold pressing** (100-400 MPa) gives a "green" compact; **degassing** removes moisture and oxides.
- **Consolidation**: hot pressing, vacuum sintering, hot isostatic pressing (HIP) or extrusion to near full density.
- **Advantages**: fine microstructure, high V_f, no chemical reaction, near-net shape. **Limitations**: costly powder, contamination (oxide films), size limit, residual porosity, safety (fine Al powder).

5.2 Liquid metallurgy techniques

Reinforcement is put into **molten** matrix; cheapest and most scalable route (casting).

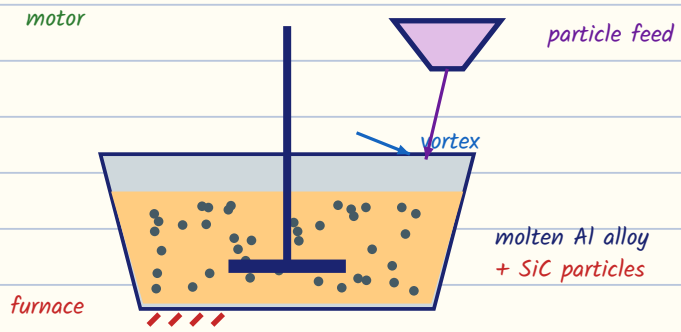


Fig. 4 Stir casting (vortex method)

- **Stir casting (vortex)**: preheated SiC (≈ 900 °C) is fed into a **vortex** created by a mechanical stirrer in molten Al (≈ 700 °C) → cast in a mould. Mg is added to improve wettability. Up to 30 % V_p; problems: particle settling, clustering, porosity (argon degassing).
- **Compocasting / rheocasting**: particles are added to **semi-solid (partially solidified) slurry**: viscosity holds particles in suspension, reaction is minimal.
- **Squeeze casting / pressure infiltration**: molten metal forced by a **hydraulic punch (50–150 MPa)** into a **preform** of fibres / particles; no porosity, high V_f (up to 60 %), near-net shape.

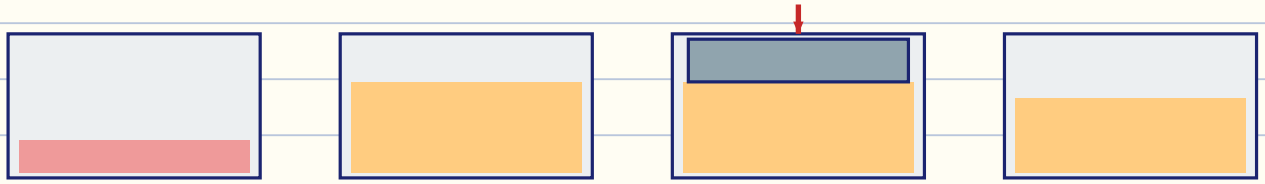


Fig. 5 Squeeze-casting infiltration

- **Gas-pressure infiltration**: argon pressure forces the melt into a preform (large parts, e.g. Al-SiC substrates); **vacuum infiltration** relies on capillarity + vacuum.
- **Spray deposition (Osprey process)**: molten alloy is **atomised** by inert gas, and reinforcement particles are injected into the spray; droplets + particles deposit on a substrate as a near-dense preform; rapid solidification gives fine grains.
- **In-situ (reactive) processing**: the reinforcement is **formed in the melt** by chemical reaction (e.g. Al + K_2TiF_6 + KBF_4 → TiB_2 particles; XD™ process); clean, well-bonded interface.

5.3 Secondary processing

- **Extrusion, rolling, forging, drawing** at 400–500 °C: break clusters, align whiskers / short fibres, close pores, improve strength and ductility.

- **Heat treatment** (solution + ageing, T6): precipitation hardening of the Al matrix is accelerated by dislocations around particles.
- **Machining, joining**: diamond / PCD tools, EDM, laser, abrasive water-jet; friction-stir / diffusion welding.
- **Superplastic forming** of fine-grained MMC sheets; HIP for densification of castings.

5.4 Special fabrication techniques

Technique	Principle	Used for
Diffusion bonding	Alternate layers of matrix foil and fibre mats (or coated fibres) are stacked and pressed 20-100 MPa at 0.6-0.8 T_m for hours; atoms diffuse across the interfaces (foil-fibre-foil)	Ti / SiC, Al / B sheets, plates
Vapour deposition (PVD / CVD / plasma spray)	Matrix metal is vapour-deposited or plasma-sprayed on to fibres (monotape), then hot-pressed	Fibre MMC tapes, turbine parts
Electrodeposition / electroforming	Co-deposition of particles / fibres while metal (Ni, Cu) is plated from the bath	Coatings, Ni-SiC wear surfaces
Mechanical alloying	High-energy ball milling repeatedly fractures and cold-welds powders → nano-dispersion	ODS alloys, nano MMC
Additive manufacturing / laser cladding	Powder or wire mixed with reinforcement fused layer by layer	Complex, graded MMCs
Friction stir processing	Rotating tool stirs reinforcement into a solid-state surface	Surface composites

Process	V_f	Cost	Reaction risk	Product
Stir casting	≤ 30 %	Low	Moderate	Large castings
Squeeze casting	≤ 60 %	Medium	Low	Near-net parts
Powder metallurgy	≤ 50 %	High	Very low	Small, high quality
Spray deposition	≤ 20 %	Medium-high	Low	Billets
Diffusion bonding	≤ 80 %	Very high	Low-medium	Fibre sheets

VI. Properties of MMCs

6.1 Physical properties

- **Density** by rule of mixtures: $\rho_c = \rho_m V_m + \rho_r V_r$ - e.g. Al + 20 % SiC → 2.8 g/cc (4 % heavier than Al).

- **Coefficient of thermal expansion (CTE)** decreases with V_r (Turner / Kerner models) and can be tailored: Al-SiC 55-65 % → 6.5-8 $\times 10^{-6}$/K ≈ chip & ceramic substrates.
- **Thermal conductivity** is high (Al-SiC 170-200 W/mK, Cu-graphite up to 500); electrical conductivity decreases with V_r.
- **Damping capacity** is higher than monolithic metal (interface, dislocation damping).

6.2 Mechanical properties

Upper bound: $E_c = E_m V_m + E_r V_r$ **Lower bound:** $1/E_c = V_m/E_m + V_r/E_r$

- **Modulus, yield & tensile strength increase** with V_r (Al-20 % SiC: E ≈ 100-115 GPa vs 70 GPa for Al; σ_y up 30-60 %); strength mechanisms: load transfer, Orowan strengthening, grain refinement (Hall-Petch), dislocation density from CTE mismatch, work hardening.
- **Ductility and fracture toughness drop** (elongation from 15 % to 1-5 %; K_{Ic} 40 → 15-25 MPa√m).
- **Fatigue strength** improves (crack initiation delayed) but crack growth is faster at high ΔK; **creep resistance** much better (particles pin dislocations, grain boundaries).
- **Stiffness in fibre direction**: continuous B/Al: E_l = 220 GPa, σ_l = 1.4 GPa at V_f = 0.5.

6.3 Wear properties

- Hard ceramic particles **carry the load and resist abrasion** - wear rate falls with V_r (Al-20 % SiC wears about 5-10 times slower than Al alloy).
- Adhesive wear and **seizure** are reduced; graphite particles act as a **solid lubricant** (Al-Gr bearings, cylinder liners).
- Mechanically mixed layer (MML) of oxides and debris forms on the worn surface and protects the sliding surface; large, hard particles are better against abrasion, small particles for adhesion.
- Counter-face wear can increase (hard particles abrade the mating steel) - use compatible materials.

6.4 Machinability

- **Poor machinability**: hard abrasive ceramic particles wear HSS / carbide tools rapidly (flank wear); built-up edge is smaller, surface finish rough due to particle pull-out.
- Use **PCD (polycrystalline diamond)** or CVD diamond tools, low feed, high speed (≈ 300-600 m/min), liquid coolant.
- Non-conventional methods: **EDM**, wire-EDM, laser, abrasive water-jet and ultrasonic machining; near-net-shape casting to minimise machining.

- Wear depends on size and V_r of the particle: tool life falls by 10-50 times for 20-30 % SiC.

6.5 Other properties

- **Corrosion**: galvanic coupling between C / SiC and Al can raise pitting; surface anodising and sealing needed.
- **High-temperature stability**: Al-SiC useful to 300 °C; Ti-SiCf to 650 °C. **Dimensional stability** and low outgassing for space.
- Good **electromagnetic shielding** and **non-magnetic** behaviour (Al, Mg MMCs).

VII. Effect of Size, Shape and Distribution of Particulate on Properties

Factor	Effect on properties
Particle size	Smaller particles (< 10 µm) → higher yield and tensile strength (Orowan / Hall-Petch, more interfaces), better ductility and toughness; large particles (> 30 µm) crack easily and promote void nucleation, but give better wear / abrasion resistance
Volume fraction	Modulus, strength, hardness, wear resistance and thermal stability ↑; ductility and toughness ↓; castability and machinability worsen above ≈ 25-30 %
Shape	Angular / sharp particles → high local stress concentration, lower ductility, but stronger interlocking. Spherical → better flow, higher ductility. High aspect ratio (whiskers, fibres) → much higher strength / modulus (load transfer) but anisotropic
Distribution	Uniform → best properties. **Clustering / agglomeration** and particle-free zones → premature failure, porosity, scatter and low ductility. Segregation by settling in casting (density difference), lessened by stirring, compocasting, extrusion
Interface	Good bonding → load transfer; excessive reaction layer (Al_4C_3) → brittle; weak interface → debonding

Concept - particle strengthening

> Orowan: $\Delta\sigma_y = 0.13\ G\ b / \lambda \cdot \ln(r/b)$ → **smaller spacing λ = stronger**.

> Shear-lag (load transfer): $\Delta\sigma = \sigma_m\ V_r\ (S + 4)/4$ for aspect ratio $S = L/d$.

> CTE mismatch: $\Delta\alpha\ \Delta T$ creates dislocations → more strengthening; also residual stresses.

VIII. Numerical Problems on MMCs

Problem 1

Al-20 vol % SiC particulate MMC: $\rho_{Al} = 2.7$, $\rho_{SiC} = 3.2$ g/cc, $E_{Al} = 70$, $E_{SiC} = 450$ GPa. Find ρ_c and the upper / lower bounds of E_c.

ρ_c = 2.7(0.8) + 3.2(0.2) = **2.80 g/cc**

Upper: E_c = 70(0.8) + 450(0.2) = 56 + 90 = **146 GPa**

Lower: $1/E_c$ = 0.8/70 + 0.2/450 = 0.011429 + 0.000444 → E_c = **84.2 GPa** (test value ≈ 100–115 GPa lies between the bounds).

Specific modulus: Al 70/2.7 = 26 → MMC ≈ 105/2.8 = 37.5 GPa/(g/cc): **40 % better**.

Problem 2

Aluminium-boron fibre composite with V_f = 0.5: E_f = 400 GPa, E_m = 70 GPa, σ_{fu} = 3500 MPa. Estimate E_1, E_2 and the strength along fibres (neglect matrix contribution).

E_1 = 400(0.5) + 70(0.5) = **235 GPa**. E_2 = 400 × 70 / (400 × 0.5 + 70 × 0.5) = 28000 / 235 = **119 GPa**.

$\sigma_{lu} \approx \sigma_{fu} V_f$ = 3500 × 0.5 = **1750 MPa** (+ small matrix part) → compare Al alloy ≈ 300 MPa.

IX. Quick Revision – Unit IV

Remember

- MMC = metal matrix + ceramic reinforcement: higher T, wear, conductivity, no moisture than PMC; costly, less ductile.
- Reinforcement: SiC, Al_2O_3, B_4C, C, B, W; matrix: Al, Mg, Ti, Cu, Ni, Fe.
- Fabrication: **PM** (blend-press-sinter-HIP), **stir casting**, **compocasting**, **squeeze / pressure infiltration**, **spray deposition**, in-situ, diffusion bonding, vapour deposition; secondary: extrusion, rolling, heat-treat.
- Properties: ↑ E, σ, wear, creep; ↓ ductility, toughness, machinability. Small, uniform, well-bonded particles are best.

Likely exam questions

- Classify MMCs. Discuss reinforcement materials and selection of base metals.
- Explain need for MMCs and applications.
- Describe powder metallurgy and liquid metallurgy techniques; write about secondary and special fabrication processes.
- Write a note on physical, mechanical, wear and machining properties. Effect of size, shape and distribution of particulates.

Short Answer Questions (2 marks)

Q1. What is an MMC?

Ans: Composite with a metal or alloy matrix reinforced with ceramic / metallic fibres, whiskers or particles.

Q2. Give two advantages of MMCs over PMCs.

Ans: Higher service temperature, no moisture absorption, better wear and thermal / electrical conductivity.

Q3. Which reinforcement is most common for Al MMCs?

Ans: SiC and Al_2O_3 particles.

Q4. Why is Mg added in stir casting of Al-SiC?

Ans: Mg lowers the surface tension and improves wetting of SiC by molten Al.

Q5. What is compocasting?

Ans: Stir casting in the semi-solid state so that particles are held in suspension.

Q6. What is squeeze casting?

Ans: Forcing molten metal into a preform under high pressure and solidifying under that pressure.

Q7. Purpose of degassing in powder metallurgy?

Ans: Removal of adsorbed moisture and gases before consolidation.

Q8. Name machining tools for MMC.

Ans: PCD / diamond-coated tools; EDM, laser, abrasive water jet.

Q9. How does particle size affect strength?

Ans: Smaller particles give higher strength and ductility; large particles crack and reduce toughness.

Q10. Which MMC is used for electronic packaging?

Ans: Al-SiC, Cu-W, Cu-Mo.

Glossary of Key Terms

Term	Meaning
Wettability	Ability of molten metal to spread on a ceramic surface
Interface	Boundary region between matrix and reinforcement where load is transferred
Preform	Porous skeleton of fibres / particles into which metal is infiltrated
HIP	Hot isostatic pressing: high temperature + gas pressure from all sides
ODS	Oxide dispersion strengthened alloy
CTE	Coefficient of thermal expansion
Orowan strengthening	Strengthening by dislocations bowing between hard particles
Whisker	Single-crystal short fibre of very high strength

Answer Outlines for 10-mark Questions

Powder metallurgy of MMC

> Blending of powders, sketch of steps.
> Cold compaction, degassing, sintering / hot pressing / HIP, extrusion.
> Advantages (fine structure, high V_f, no reaction) and limitations.

Stir casting and squeeze casting

> Sketch of stir casting set-up, role of vortex and Mg.
> Problems: settling, clustering, porosity, reaction.
> Squeeze casting procedure, pressure range, advantages.

Effect of particle size, shape and distribution

> Size: Orowan, Hall-Petch; ductility trend.
> Shape: angular vs spherical; whiskers (aspect ratio).
> Distribution: clusters, particle free zones; remedy by secondary working.

Practice Problems with Answers

Problem A

Cu-W composite with 60 vol % W (ρ_{Cu} = 8.9, ρ_W = 19.3 g/cc). Find density.

$\rho_c = 8.9(0.4) + 19.3(0.6) = 3.56 + 11.58 =$ **15.14 g/cc**.

Problem B

Al-30 vol % SiC (E_{Al} = 70, E_{SiC} = 450 GPa): find the upper bound modulus and the weight fraction of SiC (ρ = 2.7 / 3.2).

$E_c = 70(0.7) + 450(0.3) =$ **184 GPa** ; $\rho_c = 2.7(0.7) + 3.2(0.3) = 2.85$; $W_{SiC} = 3.2(0.3)/2.85 =$ **0.337**.